Lab 01 - Data Visualization

Your Name Goes Here

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Why do we visualize?

The `datasauRus` R package is a modern extension of Anscombe's quartet, a famous example used in introductory statistics courses. We will explore why the datasets are so famous in this lab.

In the dataset `datasaurus_dozen`, we have observations from 13 datasets, each of which contains 142 observations of two variables, x and y.

Exercise 1

The following code chunk calculates, for each of the 13 datasets, the following descriptive statistics for x and y: the mean, the standard deviation (square root of the variance), and the correlation between x and y. Run the code and comment on how these summary statistics vary across the groups (datasets).

```
summdat <- datasaurus_dozen %>%
  group_by(dataset) %>%
  summarise(
    mean_x = mean(x),
    mean_y = mean(y),
    sd_x = sd(x),
    sd_y = sd(y),
    r = cor(x, y)
  )

print(summdat, n = 13)
```

```
## # A tibble: 13 x 6
##    dataset    mean_x mean_y  sd_x  sd_y       r
##    <chr>       <dbl>  <dbl> <dbl> <dbl>   <dbl>
##  1 away         54.3   47.8  16.8  26.9 -0.0641
##  2 bullseye     54.3   47.8  16.8  26.9 -0.0686
##  3 circle       54.3   47.8  16.8  26.9 -0.0683
##  4 dino         54.3   47.8  16.8  26.9 -0.0645
##  5 dots         54.3   47.8  16.8  26.9 -0.0603
##  6 h_lines      54.3   47.8  16.8  26.9 -0.0617
##  7 high_lines   54.3   47.8  16.8  26.9 -0.0685
##  8 slant_down   54.3   47.8  16.8  26.9 -0.0690
##  9 slant_up     54.3   47.8  16.8  26.9 -0.0686
## 10 star         54.3   47.8  16.8  26.9 -0.0630
## 11 v_lines      54.3   47.8  16.8  26.9 -0.0694
## 12 wide_lines   54.3   47.8  16.8  26.9 -0.0666
## 13 x_shape      54.3   47.8  16.8  26.9 -0.0656
```

Exercise 2

Create a scatterplot of y vs x, faceted by dataset (be sure to use informative labels - nicer plots get more points!). Then discuss how the plots compare across the datasets. How does your response here compare to your response to Exercise 1? What does this say about using summary statistics and data visualizations as tools when getting to know a dataset?

Exercise 3: Global Life Expectancy

How has life expectancy from infancy changed over time around the world?

Data

The data we're using originally come from the Institute for Health Metrics and Evaluation.

Let's create a data visualization that displays how life expectancy in the US has changed over time for females, comparing it to life expectancy in India and Iran. Once we have that plot, you can change gender and countries, update colors, and format further as desired. We'll use the plot of statistics majors from our lecture as a model.

We can easily change which countries are being plotted by changing which countries the code above `filters` for. Note that the country name should be spelled and capitalized exactly the same way as it appears in the data. See the end of the file for a list of the countries in the data.

Target: Something Like Stats Majors Plot from Class

```
ggplot(degrees_long,
       aes(x = year, y = count, color = degree_type)) +
  geom_line() +
  labs(
    title = "Statistical Science Degrees Awarded by Type and Year",
    x = "Year",
    y = "Number of Degrees",
    color = "Degree Type"
  ) + scale_x_continuous(breaks = seq(2011, 2026, by = 1)) +
    theme(axis.text.x = element_text(angle = 45, hjust=1, size=8))
```

Starting to Code

The data we're using originally come from the Institute for Health Metrics and Evaluation.

```
life_wide <- readr::read_csv("lifeexp/lifeexpectancy_wide.csv")
```

```
## Rows: 408 Columns: 36
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr  (2): location, sex
## dbl (34): 1990, 1991, 1992, 1993, 1994, 1995, 1996, 1997, 1998, 1999, 2000, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
head(life_wide,10)
```

```
## # A tibble: 10 x 36
```

```
##    location sex   `1990` `1991` `1992` `1993` `1994` `1995` `1996` `1997` `1998`
##    <chr>    <chr>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>  <dbl>
##  1 Afghani~ Fema~   59.8   59.8   60.2   60.5   60.7   61.1   61.6   61.6   60.7
##  2 Afghani~ Male    58.1   57.7   58.0   58.6   58.2   59.3   60.1   59.6   58.2
##  3 Albania  Fema~   74.9   75.1   75.5   76.0   76.4   76.7   77.0   76.4   77.5
##  4 Albania  Male    69.2   69.4   69.8   70.2   70.6   70.8   70.8   68.5   71.8
##  5 Algeria  Fema~   69.9   70.5   70.9   71.3   71.6   72.0   72.4   72.7   73.0
##  6 Algeria  Male    67.0   67.4   67.8   68.1   68.3   68.6   69.3   69.6   69.8
##  7 America~ Fema~   73.2   73.1   73.0   72.9   72.8   72.7   72.6   72.5   72.4
##  8 America~ Male    66.3   66.2   66.2   66.2   66.2   66.3   66.4   66.4   66.4
##  9 Andorra  Fema~   83.1   83.2   83.3   83.5   83.6   83.7   83.8   83.9   84.0
## 10 Andorra  Male    76.5   76.7   76.9   77.0   77.2   77.3   77.5   77.6   77.8
## # i 25 more variables: `1999` <dbl>, `2000` <dbl>, `2001` <dbl>, `2002` <dbl>,
## #   `2003` <dbl>, `2004` <dbl>, `2005` <dbl>, `2006` <dbl>, `2007` <dbl>,
## #   `2008` <dbl>, `2009` <dbl>, `2010` <dbl>, `2011` <dbl>, `2012` <dbl>,
## #   `2013` <dbl>, `2014` <dbl>, `2015` <dbl>, `2016` <dbl>, `2017` <dbl>,
## #   `2018` <dbl>, `2019` <dbl>, `2020` <dbl>, `2021` <dbl>, `2022` <dbl>,
## #   `2023` <dbl>
```

Let's start thinking about draft code! Fill in the missing pieces to make your plot, and time permitting, vary the countries and gender, or add some bells and whistles to make the plot more visually appealing.

```
ggplot(life_wide,
       aes()) +
  geom_line()
```

Appendix

Below is a list of countries in the dataset:

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
##     group_rows
```

Country
Afghanistan
Albania
Algeria
American Samoa
Andorra
Angola
Antigua and Barbuda
Argentina
Armenia
Australia
Austria
Azerbaijan
Bahamas
Bahrain
Bangladesh

(continued)

Country
Barbados
Belarus
Belgium
Belize
Benin
Bermuda
Bhutan
Bolivia (Plurinational State of)
Bosnia and Herzegovina
Botswana
Brazil
Brunei Darussalam
Bulgaria
Burkina Faso
Burundi
Cabo Verde
Cambodia
Cameroon
Canada
Central African Republic
Chad
Chile
China
Colombia
Comoros
Congo
Cook Islands
Costa Rica
Cote d'Ivoire
Croatia
Cuba
Cyprus
Czechia
Democratic People's Republic of Korea
Democratic Republic of the Congo
Denmark
Djibouti
Dominica
Dominican Republic
Ecuador
Egypt
El Salvador
Equatorial Guinea
Eritrea
Estonia
Eswatini
Ethiopia

(continued)

Country
Fiji
Finland
France
Gabon
Gambia
Georgia
Germany
Ghana
Greece
Greenland
Grenada
Guam
Guatemala
Guinea
Guinea-Bissau
Guyana
Haiti
Honduras
Hungary
Iceland
India
Indonesia
Iran (Islamic Republic of)
Iraq
Ireland
Israel
Italy
Jamaica
Japan
Jordan
Kazakhstan
Kenya
Kiribati
Kuwait
Kyrgyzstan
Lao People's Democratic Republic
Latvia
Lebanon
Lesotho
Liberia
Libya
Lithuania
Luxembourg
Madagascar
Malawi
Malaysia
Maldives
Mali

(continued)

Country
Malta
Marshall Islands
Mauritania
Mauritius
Mexico
Micronesia (Federated States of)
Monaco
Mongolia
Montenegro
Morocco
Mozambique
Myanmar
Namibia
Nauru
Nepal
Netherlands
New Zealand
Nicaragua
Niger
Nigeria
Niue
North Macedonia
Northern Mariana Islands
Norway
Oman
Pakistan
Palau
Palestine
Panama
Papua New Guinea
Paraguay
Peru
Philippines
Poland
Portugal
Puerto Rico
Qatar
Republic of Korea
Republic of Moldova
Romania
Russian Federation
Rwanda
Saint Kitts and Nevis
Saint Lucia
Saint Vincent and the Grenadines
Samoa
San Marino

(continued)

Country
Sao Tome and Principe
Saudi Arabia
Senegal
Serbia
Seychelles
Sierra Leone
Singapore
Slovakia
Slovenia
Solomon Islands
Somalia
South Africa
South Sudan
Spain
Sri Lanka
Sudan
Suriname
Sweden
Switzerland
Syrian Arab Republic
Taiwan (Province of China)
Tajikistan
Thailand
Timor-Leste
Togo
Tokelau
Tonga
Trinidad and Tobago
Tunisia
Turkey
Turkmenistan
Tuvalu
Uganda
Ukraine
United Arab Emirates
United Kingdom
United Republic of Tanzania
United States Virgin Islands
United States of America
Uruguay
Uzbekistan
Vanuatu
Venezuela (Bolivarian Republic of)
Viet Nam
Yemen
Zambia
Zimbabwe